

A Concise Derivation of Linear Discriminant Analysis

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Goal

Given n labeled data points

$$\{(x_i, c_i)\}_{i=1}^n, \quad x_i \in \mathbb{R}^p, \quad c_i \in \{1, \dots, K\},$$

where K is the number of classes, LDA seeks a linear projection

$$y = w^T x, \quad w \in \mathbb{R}^p, \quad y \in \mathbb{R},$$

such that the projected data are **maximally separated between classes** while being **maximally compact within each class**. This single criterion is captured by the *Fisher ratio*:

$$J(w) = \frac{w^T S_B w}{w^T S_W w},$$

where S_B is the *between-class scatter matrix* and S_W is the *within-class scatter matrix* (both defined below). The derivation proceeds in four steps: define the scatter matrices, formulate the optimization, apply the generalized eigenvalue technique, and read off the solution.

Step 1: Define the Scatter Matrices

Notation.

- n_k = number of samples in class k .
- $\mu_k = \frac{1}{n_k} \sum_{i: c_i=k} x_i$ (class mean).
- $\mu = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum_{k=1}^K n_k \mu_k$ (global mean).

Within-class scatter matrix $S_W \in \mathbb{R}^{p \times p}$: measures the total spread of samples around their own class mean.

$$S_W = \sum_{k=1}^K \sum_{i: c_i=k} (x_i - \mu_k)(x_i - \mu_k)^T.$$

Between-class scatter matrix $S_B \in \mathbb{R}^{p \times p}$: measures how far the class means are from the global mean, weighted by class size.

$$S_B = \sum_{k=1}^K n_k (\mu_k - \mu)(\mu_k - \mu)^T.$$

Both S_W and S_B are symmetric positive semi-definite (see Appendix ??).

After projection onto direction w , the projected class mean is $w^T \mu_k$ and the projected global mean is $w^T \mu$. The scalar between-class and within-class scatters become, respectively,

$$\begin{aligned}\tilde{s}_B &= \sum_{k=1}^K n_k (w^T \mu_k - w^T \mu)^2 = w^T S_B w, \\ \tilde{s}_W &= \sum_{k=1}^K \sum_{i: c_i=k} (w^T x_i - w^T \mu_k)^2 = w^T S_W w.\end{aligned}$$

Step 2: Formulate the Optimization Problem

We wish to find w that maximizes class separation relative to within-class spread. Formally:

$$\max_{w \in \mathbb{R}^p} J(w) = \frac{w^T S_B w}{w^T S_W w}.$$

Why the ratio? Maximizing only $w^T S_B w$ is meaningless because we can make it arbitrarily large by scaling w . Dividing by $w^T S_W w$ removes the scale ambiguity and measures *relative* separation.

Scale invariance. $J(\alpha w) = J(w)$ for any $\alpha \neq 0$, so we may impose the normalization

$$w^T S_W w = 1$$

without loss of generality. The problem becomes an equivalent constrained optimization:

$$\max_w w^T S_B w \quad \text{subject to} \quad w^T S_W w = 1.$$

Step 3: Apply the Lagrange Multiplier Method

Using the Lagrange multiplier method (see Appendix ??), we introduce $\lambda \in \mathbb{R}$ and form the Lagrangian:

$$\mathcal{L}(w, \lambda) = w^T S_B w - \lambda (w^T S_W w - 1).$$

Taking the derivative with respect to w and setting it to zero (see Appendix ??):

$$\frac{\partial \mathcal{L}}{\partial w} = 2S_B w - 2\lambda S_W w = 0.$$

This gives the **generalized eigenvalue equation**:

$$\boxed{S_B w = \lambda S_W w.}$$

Why this is useful. The stationary points of $J(w)$ are exactly the generalized eigenvectors of the matrix pair (S_B, S_W) . Substituting a solution w back into J :

$$J(w) = \frac{w^T S_B w}{w^T S_W w} = \frac{w^T (\lambda S_W w)}{w^T S_W w} = \lambda.$$

Hence, the maximum of $J(w)$ equals the largest generalized eigenvalue $\lambda_{\max}$, achieved by the corresponding generalized eigenvector.

Step 4: Reduce to a Standard Eigenvalue Problem

If S_W is invertible (i.e., $n > p$ and no class is degenerate), multiply both sides of the generalized eigenvalue equation by S_W^{-1} :

$$S_W^{-1} S_B w = \lambda w.$$

This is a **standard eigenvalue problem** for the matrix $S_W^{-1} S_B$.

Observation on rank. S_B is a sum of K rank-1 matrices $(\mu_k - \mu)(\mu_k - \mu)^T$, so

$$\text{rank}(S_B) \leq K - 1.$$

Therefore $S_W^{-1} S_B$ has at most $K - 1$ non-zero eigenvalues, and LDA can produce **at most** $K - 1$ **discriminant directions**.

Step 5: Extract the Discriminant Directions

Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{K-1} > 0$ be the non-zero eigenvalues of $S_W^{-1} S_B$, with corresponding eigenvectors $w_1, w_2, \dots, w_{K-1}$.

To project into a d -dimensional subspace ($d \leq K - 1$), collect the top- d eigenvectors:

$$W_d = [w_1 \mid w_2 \mid \dots \mid w_d] \in \mathbb{R}^{p \times d}.$$

The low-dimensional representation of a data point x is:

$$y = W_d^T x \in \mathbb{R}^d.$$

Each eigenvector w_j is a **linear discriminant**: the direction along which the Fisher ratio is the j -th largest.

Conclusion

LDA seeks a linear projection that maximizes the Fisher ratio

$$J(w) = \frac{w^T S_B w}{w^T S_W w},$$

balancing between-class separation S_B against within-class compactness S_W . By imposing $w^T S_W w = 1$ and applying the Lagrange multiplier method, the optimality condition reduces to the generalized eigenvalue problem

$$S_B w = \lambda S_W w,$$

equivalently $S_W^{-1} S_B w = \lambda w$ when S_W is invertible. The optimal directions are the eigenvectors corresponding to the d largest eigenvalues.

LDA finds the linear directions that **maximize between-class scatter relative to within-class scatter**, obtained as the leading eigenvectors of $S_W^{-1} S_B$. At most $K - 1$ such directions exist for K classes.

Appendices

A Lagrange Multiplier Method

Given a constrained optimization problem

$$\max_w f(w) \quad \text{subject to} \quad g(w) = 0,$$

the *Lagrange multiplier method* asserts that at any local optimum w^* , there exists a scalar λ (the *Lagrange multiplier*) such that

$$\nabla_w f(w^*) = \lambda \nabla_w g(w^*),$$

i.e., the gradients of the objective and constraint are parallel. Equivalently, w^* is a stationary point of the *Lagrangian*

$$\mathcal{L}(w, \lambda) = f(w) - \lambda g(w).$$

Setting $\nabla_w \mathcal{L} = 0$ gives the necessary first-order conditions.

B Matrix Derivative of a Quadratic Form

For a symmetric matrix $A \in \mathbb{R}^{p \times p}$ and vector $w \in \mathbb{R}^p$,

$$\frac{\partial}{\partial w} (w^T A w) = 2A w.$$

Proof sketch. Writing $(w^T A w) = \sum_{i,j} A_{ij} w_i w_j$, differentiate with respect to w_k :

$$\frac{\partial}{\partial w_k} \sum_{i,j} A_{ij} w_i w_j = \sum_j A_{kj} w_j + \sum_i A_{ik} w_i = (Aw)_k + (A^T w)_k = 2(Aw)_k,$$

using symmetry $A = A^T$. Hence $\nabla_w (w^T A w) = 2Aw$.

C Symmetric Positive Semi-Definite Matrices

A matrix $M \in \mathbb{R}^{p \times p}$ is *symmetric positive semi-definite* (PSD), written $M \succeq 0$, if:

1. $M = M^T$ (symmetry), and
2. $v^T M v \geq 0$ for all $v \in \mathbb{R}^p$.

Any matrix of the form $M = \sum_i z_i z_i^T$ (a sum of outer products) is PSD, because

$$v^T M v = \sum_i v^T z_i z_i^T v = \sum_i (z_i^T v)^2 \geq 0.$$

Both S_W and S_B have exactly this form, so they are PSD.

If additionally $v^T M v > 0$ for all $v \neq 0$, then M is *positive definite* (PD), written $M \succ 0$, and M is invertible. $S_W \succ 0$ whenever the within-class data span $\mathbb{R}^p$ (i.e., $n > p$ with full-rank scatter), which is the condition required in Step 4 for S_W^{-1} to exist.

D Generalized Eigenvalue Problem

Given two symmetric matrices $A, B \in \mathbb{R}^{p \times p}$ with $B \succ 0$, the *generalized eigenvalue problem* is

$$A v = \lambda B v.$$

A scalar λ satisfying this (for some $v \neq 0$) is a *generalized eigenvalue*; the corresponding v is a *generalized eigenvector*. When B is invertible, this reduces to the standard eigenvalue problem $B^{-1}A v = \lambda v$. The generalized eigenvectors diagonalize the ratio $v^T A v / v^T B v$, and the maximum of this ratio equals the largest generalized eigenvalue.